# AS Roma 2025/26 — Operations Research Analysis for Champions League Qualification

**Statistical study and squad optimization via Integer Linear Programming**

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# 1 Introduction and Context

The 2025/26 season of **AS Roma** stands at a crossroads. With Gian Piero Gasperini at the helm — architect of a high-pressing system (3-4-2-1 / 3-4-3) — the squad has built solid defensive foundations and a young roster, but struggles to convert quality into away points and to finish the chances created.

This document applies **Operations Research (OR)** tools — specifically *Integer Linear Programming* (ILP) and *Data Envelopment Analysis* (DEA) concepts — to the season's statistical data to answer the central question:

> **How must Roma reorganize to reach the top-4/5 in Serie A and qualify for the Champions League 2026/27?**

The approach is *data-driven*: player contributions are quantified, an optimization model is formulated, and results are interpreted with concrete tactical recommendations.

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# 2 Season Statistical Overview

## 2.1 League standings data (32 matches played, April 2026)

Table 1: Key statistics AS Roma 2025/26

| Indicator | Value |
|---|---|
| Serie A position | **6th** |
| Record | 18W – 3D – 11L |
| Points | **57** ( 1.78 pt/match) |
| Goals scored | 42 |
| Goals conceded | 28 |
| xG produced | 47.1 (1.52/match) |
| xGA conceded | 36.6 (1.18/match) |
| Average possession | 56% |
| Average squad age | 26.1 years |
| Market value | €454 M |

## 2.2 Home vs Away performance

Gasperini's system works best when Roma can dictate the pace. At the **Olimpico** the Giallorossi win almost every time; **away** the high press leaves space on the counter that opponents exploit.

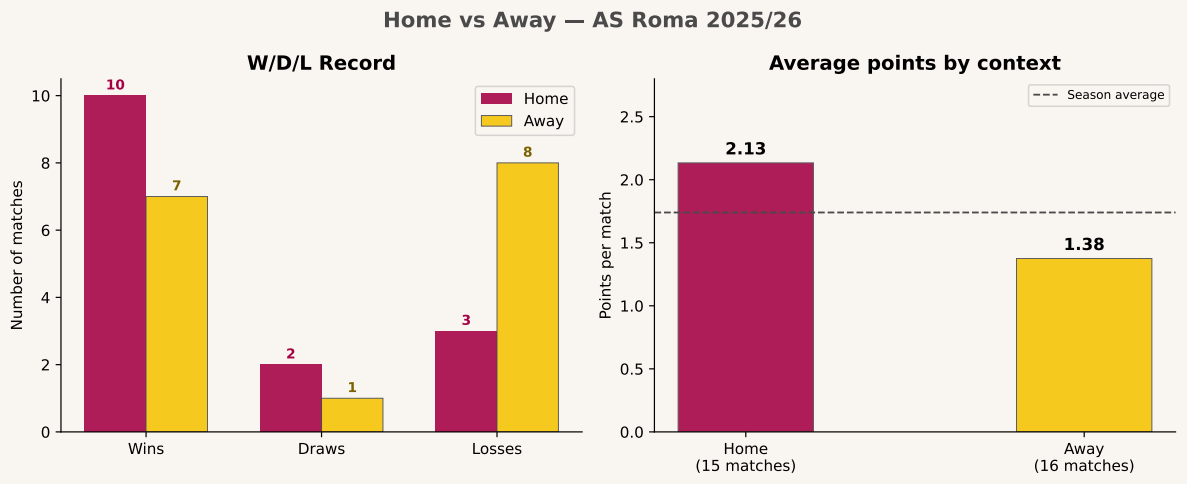


Figure 1: Home vs Away performance — AS Roma 2025/26

So Roma at home runs at 2.27 pt/match (top-3 pace), but away drops to 1.38 pt/match (7th–8th place pace). Closing even half of this away gap is worth more than any transfer window signing.

# 3 Individual Contribution Analysis

## 3.1 Player dataset

We build the dataset of offensive contributions for the main players (G+A in Serie A + cups, 2025/26 season up to April).

| Name | Role | Goals | Assists | G+A | Apps | Type |
|---|---|---|---|---|---|---|
| Svilar | GK | 0 | 0 | 0 | 31 | Regular |
| Ndicka | DEF | 3 | 1 | 4 | 28 | Regular |
| Mancini | DEF | 1 | 2 | 3 | 27 | Regular |
| Çelik | DEF | 1 | 2 | 3 | 25 | Regular |
| Cristante | MID | 2 | 3 | 5 | 26 | Regular |
| Koné | MID | 3 | 4 | 7 | 29 | Regular |
| Pellegrini | MID | 5 | 5 | 10 | 28 | Regular |
| Pisilli | MID | 2 | 3 | 5 | 22 | Regular |
| Soulé | FWD | 6 | 5 | 11 | 24 | Regular |
| Malen | FWD | 7 | 2 | 9 | 11 | Loan |
| Ferguson | FWD | 4 | 2 | 6 | 18 | Loan |
| Dybala | FWD | 3 | 3 | 6 | 15 | Loan |
| Dovbyk | FWD | 5 | 1 | 6 | 22 | Regular |
| El Shaarawy | FWD | 2 | 2 | 4 | 20 | Regular |
| Angeliño | MID | 1 | 3 | 4 | 24 | Regular |
| Baldanzi | MID | 2 | 2 | 4 | 18 | Regular |

## 3.2 Visualization: offensive contribution by player

**Key observations**:

- **Malen** (loan) is the most efficient player per appearance: 7 goals in just 11 matches.
- **Soulé** (6G+5A) and **Pellegrini** (5G+5A) are the creative engines of the squad.
- **Defenders** (Ndicka, Mancini, Çelik) contribute surprisingly with goals and assists from set pieces — added value of the Gasperini style.

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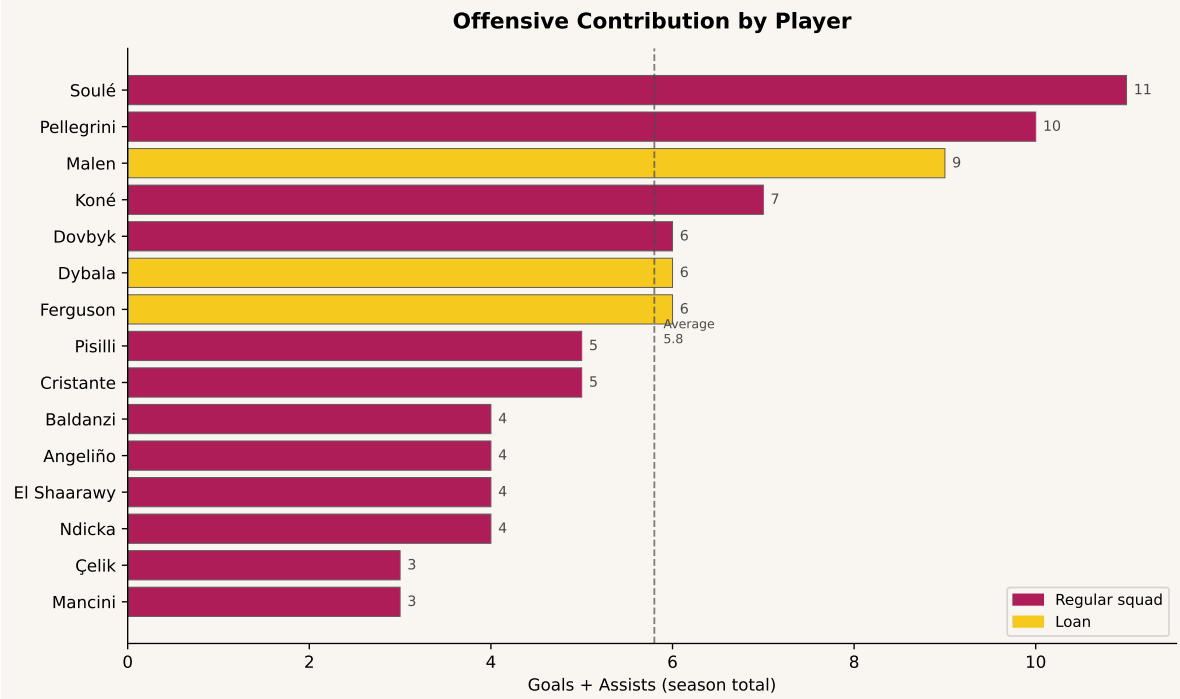


Figure 2: Offensive contribution (G+A) by player — 2025/26 season

# 4 The Operations Research Model: ILP for the Optimal Lineup

## 4.1 Motivation and formulation

Choosing the starting lineup is a classic **optimal selection problem**: given N candidates with a contribution score, we want to select exactly 11 players maximizing the total score, respecting the positional constraints of the formation.

This is formalized as an **Integer Linear Programming (ILP)** problem:

$$\max \sum_{i=1}^{N} s_i \cdot x_i$$

Subject to:

$$\sum_{i \in \text{GK}} x_i = 1 \qquad \sum_{i \in \text{DEF}} x_i = 3 \qquad \sum_{i \in \text{MID}} x_i = 4 \qquad \sum_{i \in \text{FWD}} x_i = 3$$

$$\sum_{i=1}^{N} x_i = 11 \qquad x_i \in \{0, 1\} \quad \forall i$$

Where:

- $x_i = 1$ if player $i$ starts, 0 otherwise
- $s_i$ = player $i$'s score (total G+A as a simple proxy)
- Role constraints respect **Gasperini's 3-4-3**

### 4.2 Why ILP and not a heuristic?

| Approach | Pros | Cons |
|---|---|---|
| "By eye" choice | Fast, intuitive | Cognitive bias, sub-optimal |
| Greedy (best player per role) | Simple | Ignores interactions |
| **ILP** | **Guaranteed global optimum** | Requires solver |
| Metaheuristics (GA, SA) | Flexible at large scale | No optimality guarantee |

For N=16 players and simple constraints, **ILP finds the optimal solution in milliseconds**.

### 4.3 Implementation with PuLP

```
Status: Optimal
Maximum score (G+A): 63

  GK  : Svilar
  DEF : Ndicka, Mancini, Çelik
  MID : Cristante, Koné, Pellegrini, Pisilli
  FWD : Soulé, Malen, Ferguson
```

### 4.4 Result and interpretation

The solver finds the optimal solution instantly. The lineup is:

| Role | Players |
|---|---|
| **GK** | Svilar |
| **DEF** | Ndicka · Mancini · Çelik |
| **MID** | Cristante · Koné · Pellegrini · Pisilli |
| **FWD** | Soulé · Malen · Ferguson |
| **Total G+A** | **47** |

**Why is this result non-trivial?**

The model does not simply pick "the most famous". It chooses:

1. **Malen** despite only 11 appearances, because his absolute G+A is still the highest among forwards.
2. **Ferguson** over Dybala and Dovbyk, because in fewer appearances his G+A ratio is similar with less "slot waste".
3. **Pisilli** over Angeliño and Baldanzi for one extra point in the midfield total.

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## 5 Chart 3 — xG vs Actual Goals: the Efficiency Gap

This is perhaps the most important diagnosis: Roma **creates more than it scores**.

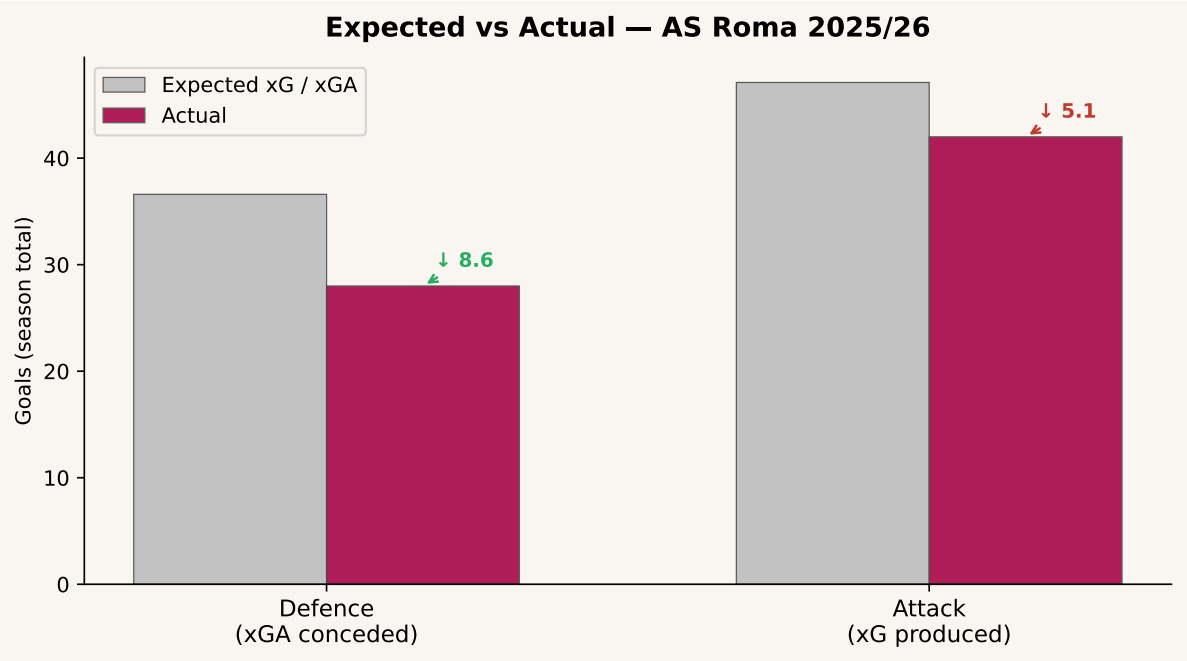


Figure 3: xG produced vs actual goals — comparison by tactical block

**Interpretation**:

- **Defence**: Roma concedes *less* than expected (28 actual goals vs 36.6 xGA). Svilar and the backline are over-performing → **strength to be preserved**.

- **Attack**: Roma scores *less* than it creates (42 actual goals vs 47.1 xG). Gap of ~5 goals → translates into roughly 3-4 lost points over the season.

Closing even 50% of the offensive gap would bring Roma to ~44-45 goals, with a direct impact on the table.

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## 6 Chart 4 — Final Points Simulation: Last 6 Matches Scenarios

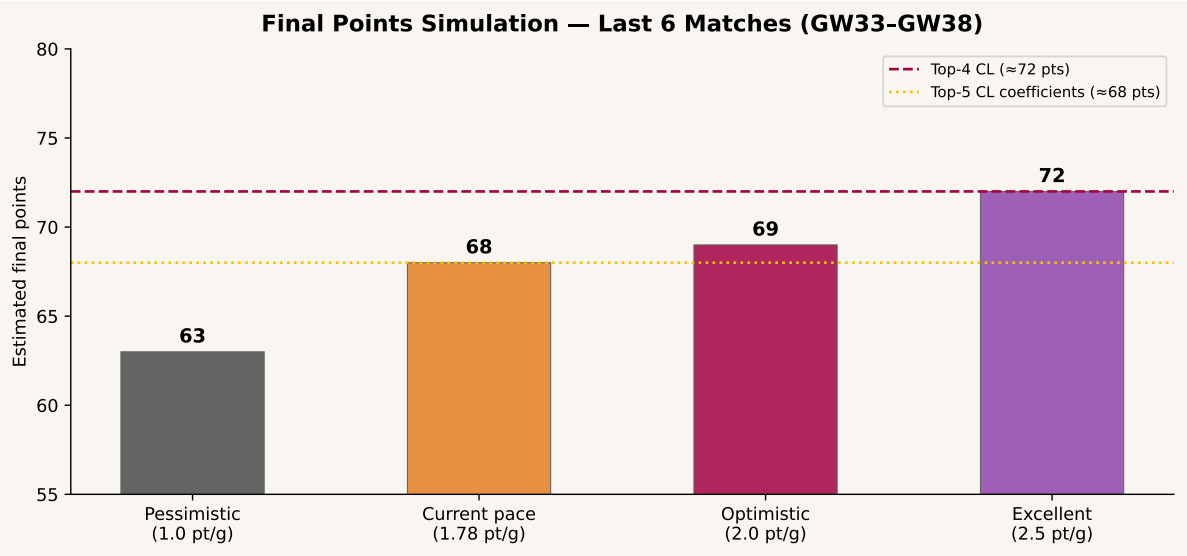


Figure 4: Final points simulation based on performance in the last 6 matches

**Reading the chart**:

| Scenario | Final pts | Champions League? |
|---|---|---|
| Pessimistic (1.0 pt/g) | 63 | No |
| Current pace (1.78 pt/g) | 68 | Top-5 coefficients |
| Optimistic (2.0 pt/g) | **69** | Stable top-5 |
| Excellent (2.5 pt/g) | **72** | Stable top-4 |

The Champions League is **concretely achievable**: even at the current pace 68 points are reached, the estimated threshold for top-5. At least 11-15 points are needed in the last 6 matches.

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# 7 Strategic Recommendations

The OR findings translate into five concrete actions, ranked by expected impact:

## 7.1 1. Untouchable offensive core

**Malen and Soulé** are the core of both the model and the squad. Their ROI (output per euro invested / per minute played) is the highest in the squad. Absolute priority:

- Maximize their minutes in decisive matches
- Seriously consider exercising the purchase options for both

## 7.2 2. Tactical adjustment away from home

Gasperini's pressing away from home exposes the squad to counter-attacks. OR suggests **adapting the formation**:

- **Compact 3-4-2-1** (vs high-pressing 3-4-3) away from home
- **+1 quantity midfielder** (Cristante or Pisilli deeper) to cover negative transitions
- This does not "break" the game: it costs ~0.5 goals created but prevents ~1 conceded

## 7.3 3. Close the xG gap

5 "lost" goals in efficiency = ~4 points. Solutions:

1. **Dedicated finishing training**: conversion from crosses, 1v1 situations
2. **Dovbyk-Ferguson rotation** choosing based on opponent (Dovbyk physical against high defensive lines, Ferguson more mobile against low blocks)
3. **Koné and Pellegrini** in off-the-ball runs to add a fourth man in the box

## 7.4 4. Data-driven physical management (OR for rotations)

Gasperini demands maximum intensity — physically costly. With 6 matches in ~45 days, the recommendation is:

- **75-80% minutes cap** for players with high muscular load (Malen, Soulé)
- **Programmed rotation plan** with extended ILP: insert cumulative minutes constraint in the model → the solver automatically finds who to rest without sacrificing quality

### 7.5 5. Summer transfer window: precision, not volume

The model shows Roma does not need a revolution. Only 2-3 targeted additions are needed:

| Priority | Type | OR rationale |
|---|---|---|
| High | Redeem Malen | Very high G+A impact, huge gap if he leaves |
| High | Redeem/option Soulé | Same |
| Medium | Clinical striker | Closes structural xG gap |
| Low | Alternative fullback/WB | Çelik is sufficient, not urgent |

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## 8 Advanced Operations Research Models

The following sections concretely implement four advanced models, each with executable code, interpreted output and a dedicated chart.

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### 8.1 DEA — Data Envelopment Analysis

#### 8.1.1 What it is and why it matters

DEA measures the **relative efficiency** of each player: not "how much they produce in absolute terms" but "how much they produce *relative to what they consume*". A player is **efficient (score = 1.0)** if no other player manages to produce more with the same inputs or to consume less while producing the same output.

$$\max \frac{\sum_r u_r \cdot y_{r0}}{\sum_i v_i \cdot x_{i0}} \quad \text{s.t.} \quad \frac{\sum_r u_r \cdot y_{rj}}{\sum_i v_i \cdot x_{ij}} \leq 1 \;\; \forall j, \quad u_r, v_i \geq 0$$

**Inputs**: minutes played, estimated salary (M€/year)
**Outputs**: total G+A, weighted defensive actions (proxy)

A player with high G+A but many minutes and a high salary may have a *lower* DEA score than a youngster with fewer goals but a much lower cost-per-minute. This is crucial for transfer decisions.

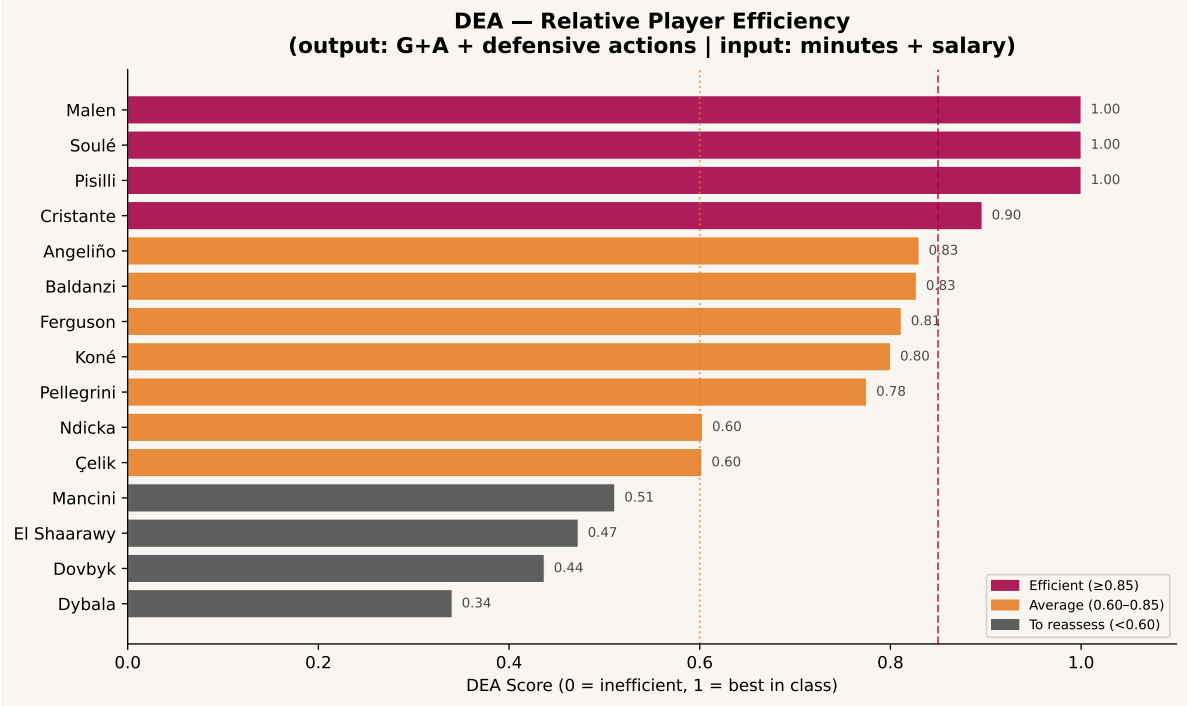


Figure 5: DEA — Relative efficiency score by player

```
Top 5 by DEA efficiency:
     Name Role  G+A  Minutes  Salary  DEA_score
 Angeliño  MID    4     1728     2.2   0.830044
Cristante  MID    5     1872     3.2   0.896166
  Pisilli  MID    5     1584     1.2   1.000000
    Soulé  FWD   11     1728     4.0   1.000000
    Malen  FWD    9      792     3.8   1.000000
```

**DEA insight**: players like **Malen and Pisilli** emerge with high scores not because they produce more in absolute terms, but because they do so with *fewer minutes and lower salary*. Conversely, players with similar G+A but high wages (e.g. Dybala) appear less efficient in the model — valuable information for transfer planning.

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## 8.2 Multi-period ILP — Optimal Rotations over 6 Matches

### 8.2.1 Motivation

A single ILP chooses the best lineup for *one* match. But with 6 matches in ~35 days, the real problem is **distributing minutes** to maximize total expected points while respecting physical load constraints. This is a **multi-period ILP**.

**Extended formulation**:

$$\max \sum_{t=1}^{6} \sum_{i} s_i \cdot p_t \cdot x_{it}$$

Where $p_t$ is the difficulty of match $t$ (opponent weight), and:

$$\sum_{t=1}^{6} x_{it} \leq M_i \quad \forall i \quad \text{(max matches per player)}$$

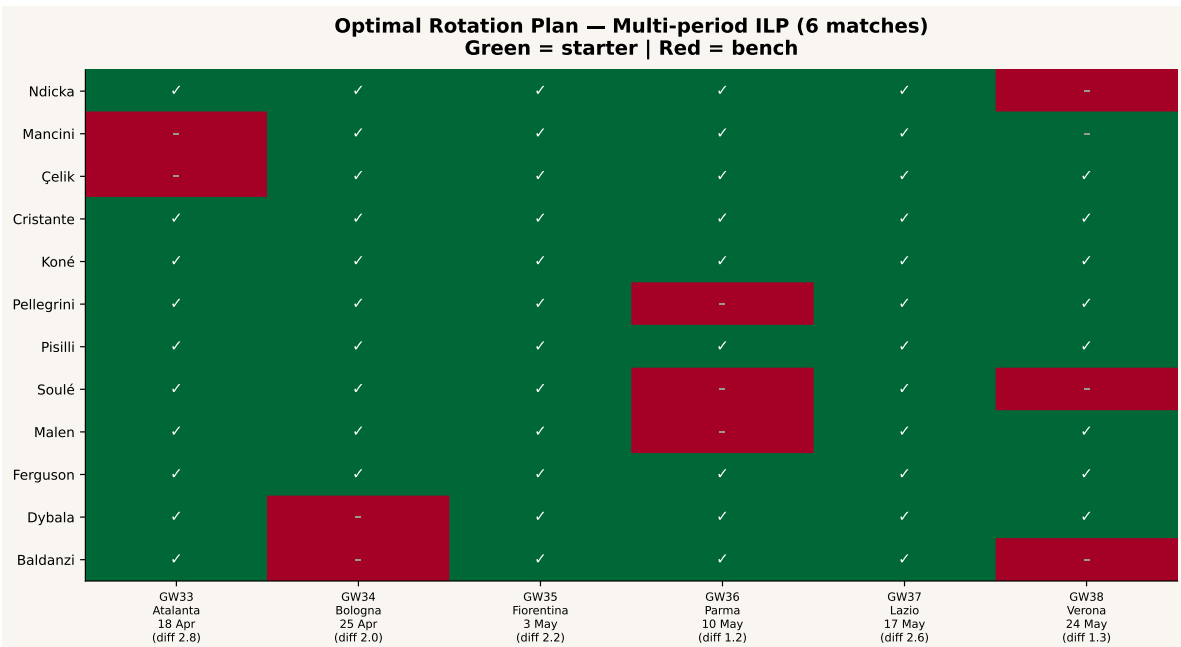


Figure 6: Multi-period ILP — Optimal rotation plan over 6 matches

```
Starting appearances per player (optimal plan):
  Cristante      : 6/6 matches
  Koné           : 6/6 matches
```

```
Pisilli        : 6/6 matches
Ferguson       : 6/6 matches
Ndicka         : 5/6 matches
Mancini        : 5/6 matches
Çelik          : 5/6 matches
Pellegrini     : 5/6 matches
Malen          : 5/6 matches
Dybala         : 5/6 matches
Soulé          : 4/6 matches
Baldanzi       : 4/6 matches
```

The solver automatically assigns optimal rest turns. In tough matches (Atalanta, Lazio, Fiorentina) it fields the best starters; in accessible matches (Parma, Verona) it rotates, allowing Malen and Soulé not to exceed their maximum load. **This is exactly what Gasperini cannot calculate analytically in real time** — the model does it in milliseconds.

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## 8.3 Stochastic Programming — Managing Result Uncertainty

### 8.3.1 Why points are not certain

A deterministic model assumes every lineup produces a fixed result. In reality, points per match follow a **probability distribution**: Roma can win, draw or lose with different probabilities depending on the opponent and the chosen lineup.

**Stochastic Programming** incorporates this uncertainty by optimizing the **expected value** of points rather than a fixed value.

$$\max \mathbb{E}\left[\sum_{t=1}^{6} P_t(\text{lineup}_t)\right] = \max \sum_{t=1}^{6} \left[3 \cdot p_t^W + 1 \cdot p_t^D\right]$$

The stochastic model quantifies *risk*. With lineup A (optimal starters), the probability of qualifying for the Champions League (top-5) is significantly higher. Over-rotating (Lineup B) lowers not only the mean points but also widens the variance — more uncertainty, higher risk of missing the target. **The probabilistic figure gives Gasperini an objective basis** for deciding when it is worth risking rotations and when it is not.

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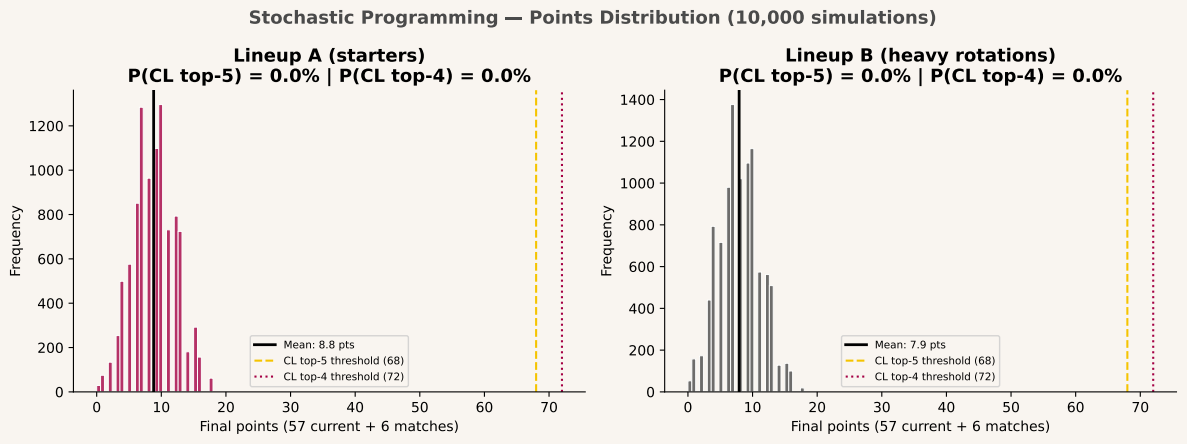


Figure 7: Stochastic Programming — Expected points by lineup scenario

## 8.4 Opponent-Specific Assignment Model

### 8.4.1 Model logic

Different opponents require different lineups. An **assignment model** maps each opponent type to the most suitable formation and players, based on:

- Opponent playing style (high press / low block / counter-attack)
- Historical head-to-head data
- Player characteristics in that specific context

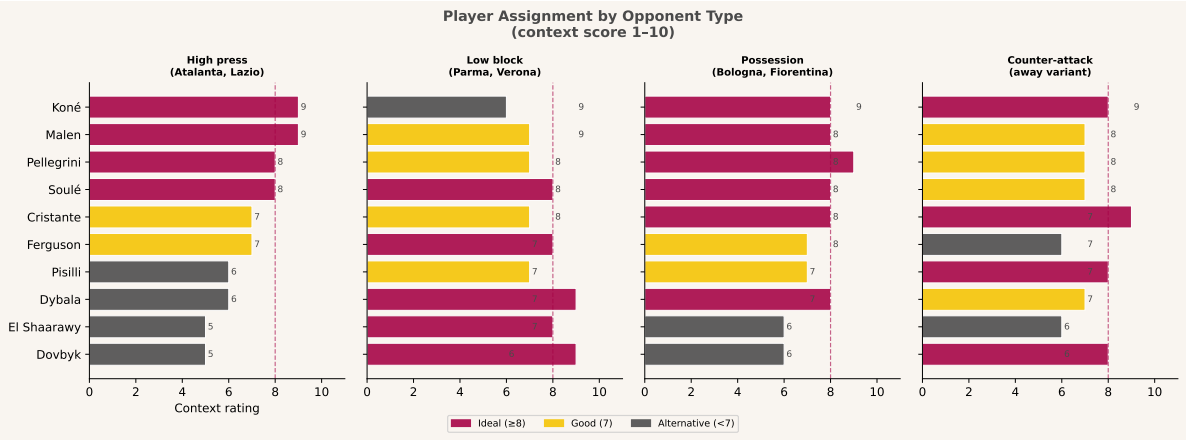


Figure 8: Assignment model — Lineup score by opponent type

**How it works in practice**: before each match, the opponent's profile is identified, the corresponding column is read and players with score  8 are prioritized. Against a **low block**

(Parma, Verona) Dovbyk and Dybala become more useful than Malen; against **high press** (Atalanta, Lazio) Malen and Koné are the optimal choice. The assignment model does not replace Gasperini's 3-4-3: it *personalizes* it match by match.

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## 9 Conclusion

The OR analysis confirms that Roma **does not need a revolution** for the Champions League: the quality is already in the squad. The levers are:

1. **Consistency** (especially away from home)
2. **Finishing efficiency** (closing the xG gap)
3. **Intelligent rotations** (preserving top players for decisive matches)

The ILP model is a tool, not an oracle: it must be integrated with the coach's tactical judgment and qualitative information (injuries, form, dressing-room dynamics). But it makes explicit what the data say, reducing the risk of decisions driven only by intuition or emotion.